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The Influence of Different Autologous Tendons on the Recovery of Knee Muscle Strength Following Anterior Cruciate Ligament Reconstruction: A Systematic Review and Network Meta-analysis

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Abstract

Background Outcomes of anterior cruciate ligament reconstruction (ACLR) vary significantly depending on autograft selection, particularly in restoring knee flexor and extensor strength, which are essential for athletic performance. The aim of this study was to investigate the impact of ACLR with different autografts on the recovery of knee flexor and extensor muscle strength. Subgroup analyses by follow-up duration were performed to characterize recovery across postoperative phases.

Methods A frequentist network meta-analysis of randomized controlled trials (RCT) and observational studies was conducted. Databases (PubMed, Embase, Web of Science, Cochrane Library) were searched from inception to January 2025. Eligible studies reported isokinetic strength outcomes after ACLR. The primary outcome was the limb symmetry index (LSI) for peak torque, which represented the ratio of injured-limb strength to uninjured-limb strength, reflecting inter-limb symmetry. Secondary outcomes were peak torque, the hamstring-to-quadriceps (H/Q) ratio and side-to-side deficit. Risk of bias (RoB) was assessed using the Cochrane RoB and the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS). Certainty of evidence was graded using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework. Graft performance was ranked using surface under the cumulative ranking curve (SUCRA) probabilities, and mean differences (MDs) with 95% confidence intervals (CIs) were calculated.

Results Thirty-one studies involving 2384 patients were included in the quantitative analysis. For knee extensor strength, assessed using LSI at 60°/s, 4-strand semitendinosus tendon (4S-ST) and 4-strand hamstring tendon (4S-HT) autografts outperformed bone-patella tendon-bone (BPTB) and quadriceps tendon (QT) autografts. At 180°/s, the 4S-HT autograft showed higher knee extensor strength LSI than the QT autograft, significantly outperformed the BPTB autograft at long-term follow-up (≥ 18 months), and consistently ranked highest according to SUCRA. The

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BPTB autograft also showed higher knee extensor strength LSI than the QT autograft. For knee flexor strength (LSI, at 60°/s and 180°/s), the QT autograft outperformed the 4S-HT autograft, with SUCRA consistently identifying the QT autograft as the top-ranked option and the 4S-HT autograft as the lowest-ranked option. No significant intergroup LSI differences were observed at short-term follow-up (≤ 6 months), whereas QT autograft's superiority in knee flexor strength became evident in the mid-term follow-up (6–12 months). The 4S-ST autograft may be superior to 4S-HT autograft in knee flexor strength LSI. Network inconsistency tests showed no significant inconsistency. However, GRADE assessments rated the certainty of evidence for comparisons of knee extensor and flexor strength between autografts as very low to low.

Conclusions The 4S-HT autograft was associated with greater knee extensor strength LSI, whereas the QT autograft had the most consistent advantage in knee flexor strength LSI. The 4S-ST autograft ranked between these grafts, but may provide greater knee flexor strength LSI than the 4S-HT autograft. The BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft. Given the very low to low certainty of evidence, these findings should be interpreted with caution.

PROSPERO registration number CRD42024530838.

Key points

- The 4S-HT autograft was associated with greater knee extensor strength LSI.
- The QT autograft had the most consistent advantage in knee flexor strength LSI.
- The 4S-ST autograft may provide greater knee flexor strength LSI than the 4S-HT autograft.
- The BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft.
- The certainty of evidence for these comparisons was very low to low.

Keywords Anterior cruciate ligament reconstruction, Autograft, Muscle strength, Network meta-analysis

Introduction

Anterior cruciate ligament (ACL) injuries, which account for more than 50% of knee injuries [1], are common among people aged 16–39, with an incidence of approximately 85 cases per 100,000 individuals [2]. Athletes, particularly those involved in high-risk sports such as soccer, football, basketball, and lacrosse, have an increased risk of ACL injury, with more than three-quarters of such injuries requiring surgical intervention [3, 4]. ACL reconstruction (ACLR) is the standard surgical procedure for these injuries [5].

Varying degrees of muscle strength decline occur in the muscles surrounding the knee joint following ACLR [6]. Knee extensor strength remains significantly reduced at 1 year, with limited improvement up to and beyond 5 years after surgery [6]. Different autografts may have distinct effects on recovery of knee flexor and extensor muscle strength after ACLR [7–10]. Bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts are the most commonly used grafts [11, 12]. BPTB autografts have historically been the gold standard for patients with significant functional requirements due to their benefits in knee joint stability and improved return-to-sports (RTS) [13, 14]. They may also lead to better knee flexor strength recovery than HT autografts [15],

but may cause anterior knee pain and reduce knee extensor strength [15, 16]. HT autografts allow for a small incision and avoid disruption of the extensor mechanism, but may increase the risk of bone tunnel widening and infection [13]. Additionally, quadriceps tendon (QT) autografts are gaining popularity [17]. QT autografts provide a large graft diameter and can also achieve a small incision, although long-term outcome data remain limited [13]. Due to the advantages and disadvantages of each tendon autograft, controversy persists regarding the optimal choice for ACLR. Therefore, to ensure the best treatment strategy for RTS, adjustments should be made according to the specific needs of the injured athlete [13].

Safe and timely RTS is a primary goal for all athletes [18] and is one of the most frequently used endpoints for evaluating surgical techniques [19]. A systematic review by Ardern et al. [20] found that 82% of patients had returned to some form of sports participation, 63% returned to their preinjury level, and 44% had resumed competitive sports at the final follow-up. Additionally, beyond simply returning to sport, many athletes are concerned about whether they can regain their pre-injury level of sport participation and performance [21, 22]. Muscle strength is fundamental to enhancing force-time characteristics (e.g. rate of force development and

external mechanical power) [23–25], optimizing sport skills such as jumping, sprinting and changing direction [26–28], and sport-specific performance [29, 30]. As such, it is an important indicator of athletic performance [31]. In a survey by the Pediatric Research in Sports Medicine (PRiSM) Society, the most commonly used testing methods were manual muscle testing (60%), followed by isokinetic (28%) and isometric (12%) tests [32].

The isokinetic muscle strength test quantifies muscle strength at specific angular velocities and is considered the "gold standard" for evaluating muscle strength [32]. It also exhibits high reproducibility at angular velocities of 60°/s and 180°/s, making it an effective tool for assessing both knee flexor and extensor strength [33].

One recent systematic review compared isokinetic muscle strength between QT, BPTB and HT autografts at different angular velocities and follow-up times [8]. QT autografts showed superior knee flexion strength compared to HT autografts and similar results to BPTB autografts at 3, 6 and 12 months [8]. Conversely, HT autografts demonstrated superior knee extensor strength at 6 and 12 months compared with QT autografts [8]. However, that review was limited to pairwise comparisons between two graft types at a time and did not include a network analysis across multiple autografts [8]. Therefore, a network meta-analysis is required to comprehensively compare the postoperative muscle strength recovery outcomes of multiple autograft types, thereby enabling indirect comparisons among various autografts and providing treatment ranking evidence to support clinical surgical decision-making.

This present study aimed to compare the effects of different autografts on postoperative recovery level of knee flexor and extensor muscle strength after ACLR using network meta-analysis. Furthermore, by integrating findings from this network meta-analysis with existing evidence, clinicians may select the most appropriate autograft according to the specific muscle strength recovery needs of individual patients.

Methods

This frequentist network meta-analysis was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta Analyses-Network Meta Analyses (PRISMA-NMA) [34]. The protocol was registered in the International Prospective Registry of Systematic Reviews (PROSPERO, CRD42024530838).

Search Strategy

A systematic literature search was performed in PubMed, Embase, Web of Science and the Cochrane Library from database inception to January 2025. No language restrictions were applied, and grey literature was excluded.

Search terms were derived from the Medical Subject Headings (MeSH) database and related keywords for ACLR and rehabilitation. Boolean operators and database-specific filters were applied. The full search strategy is provided in Supplementary Table S1.

Inclusion Criteria

Studies were included according to the PICOS framework [35]: (1) patients of any age who had undergone primary ACLR surgery with no ligament injuries in contralateral knee; (2) interventions involving different tendon autografts for ACLR; (3) comparisons between different tendon autografts; (4) muscle strength assessments using isokinetic testing at angular velocities of 60°/s or 180°/s; and (5) RCTs or observational studies.

Exclusion Criteria

Studies were excluded if they met any of the following criteria: (1) duplicates; (2) meta-analyses or reviews; (3) studies not retrieved (those that may meet the inclusion criteria but remain undetected due to limitations in search strategies, insufficient database coverage, language barriers, and other factors); (4) incomplete outcome data (e.g. mean values without standard deviations [SDs]) or unavailable follow-up data; or (5) studies whose interventions, populations, outcomes, or measurement methods were found to not meet the inclusion criteria after reviewing the full texts.

Literature Screening

The literature screening was conducted independently by two authors, first based on the title and abstracts, followed by full-text review. Inter-rater agreement was assessed using the Cohen's kappa coefficient. Discrepancies were resolved by a third reviewer. The PRISMA 2020 flow diagram for systematic reviews [36] was used to document the study selection process.

Risk of Bias and Quality of Evidence

Risk of bias (RoB) was assessed independently by two researchers, with discrepancies resolved through discussion with a third researcher. RCTs were evaluated using the Cochrane RoB tool [37], which examines seven potential sources of bias: randomization sequence generation, allocation concealment, blinding of both patients and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting and other biases. Each domain was classified as low, high or unclear risk. Review Manager (RevMan) [Computer program]. Version 5.4. The Cochrane Collaboration, 2020 was used to facilitate a visual summary of RoB across studies. Observational studies were assessed using the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS)

[38], which evaluates six potential sources of bias: patients selection, confounding variables, measurement of exposure, blinding of outcome assessment, incomplete outcome data and selective outcome reporting. Each domain was classified as low, high or unclear risk.

The certainty of evidence for network meta-analysis estimates was evaluated using the method proposed by Izcovich et al. [39], which integrates the confidence of direct, indirect, and network estimates.

Additionally, the certainty of evidence for all outcome measures was assessed using the Grading Recommendations Assessment, Development and Evaluation (GRADE) system [40, 41]. Evidence was categorized as high, moderate, low or very low based on risk of bias, inconsistency, indirectness, imprecision and publication bias.

Data Extraction

Two researchers independently extracted relevant study characteristics, with discrepancies resolved by a third researcher. Data were collected using standardized extraction forms to ensure the validity and reliability of the results. Extracted information included first author, year of publication, study design, country, type of tendon autograft, patient demographics, number of drop-outs, duration of follow-up, and outcome measures. For pooled analyses of baseline mean age and sex distribution, the overall mean age was calculated using a sample size-weighted method, and the pooled proportion of male participants was calculated by summing the number of male participants and dividing by the total sample size. Studies with unavailable or incomplete baseline age or sex data were excluded from these demographic syntheses.

Because general patient characteristics may differ (e.g., athletes typically demonstrate greater muscle strength than the general population), peak torque values alone may be influenced by individual heterogeneity. Therefore, the limb symmetry index (LSI) for peak torque was selected as the primary outcome for assessing muscle strength recovery after ACLR with different tendon autografts. LSI was calculated as the ratio of injured-limb strength to uninjured-limb strength and reflects inter-limb strength symmetry rather than absolute strength; therefore, a higher LSI does not necessarily indicate greater absolute strength in the affected limb. Secondary outcomes included peak torque, the hamstring-to-quadriceps (H/Q) ratio and side-to-side deficit. Peak torque refers to the maximum torque generated by a muscle group during joint movement, reflecting maximal muscle strength and serving as a gold standard indicators in the isokinetic test [42]. The H/Q ratio was defined as the ratio of knee flexion strength to extension strength,

while side-to-side deficit was calculated as the difference between non-injured and injured limb torque, divided by the non-injured limb torque.

Network Meta-analysis Methodology

Network meta-analysis is an extension of traditional pairwise meta-analysis that allows indirect comparisons between interventions not directly compared in the included studies. Data were extracted from study tables or text, including sample size, means, and SD. The “network” suite of commands in Stata (USA, Texas, 18.0) was utilized to generate network plots for each outcome, illustrating the autograft types compared across studies. Inconsistency was assessed using the inconsistency model to evaluate global inconsistency. If global inconsistency was detected, a node-splitting approach was applied to assess local inconsistency in closed-loop structures. Forest plots of mean differences (MDs), reported together with their corresponding 95% confidence intervals (CIs), were constructed to visualize individual study results and pooled estimates. Autograft types were ranked using the surface under the cumulative ranking (SUCRA), and ranking tables were generated to compare the relative effect differences among the autograft types. Funnel plots were employed to assess potential publication bias.

Postoperative recovery of knee extensor and flexor strength after ACLR is time dependent, with rapid improvement up to 12–18 months and minimal gains thereafter [6]. Accordingly, subgroup analyses were performed based on follow-up duration: short-term (≤ 6 months), mid-term (6–12 months), and long-term (≥ 18 months). No studies reported follow-up between 12 and 18 months, so this interval was not analyzed. A two-sided P -value < 0.05 was considered statistically significant.

Results

Literature Selection

A comprehensive search of databases identified 1434 studies. Following strict inclusion and exclusion criteria, a total of 31 studies were finally included. The inter-rater agreement between the two reviewers during literature selection was excellent, with a Cohen's kappa coefficient of 0.966. The screening process is shown in Fig. 1, and the specific reasons for exclusion are presented in Supplementary Table S3.

General Information and Characteristics of Included Studies

The general characteristics of the 31 included studies are summarized in Supplementary Table S4. Of these, eleven were RCTs [43–53] and 20 were observational

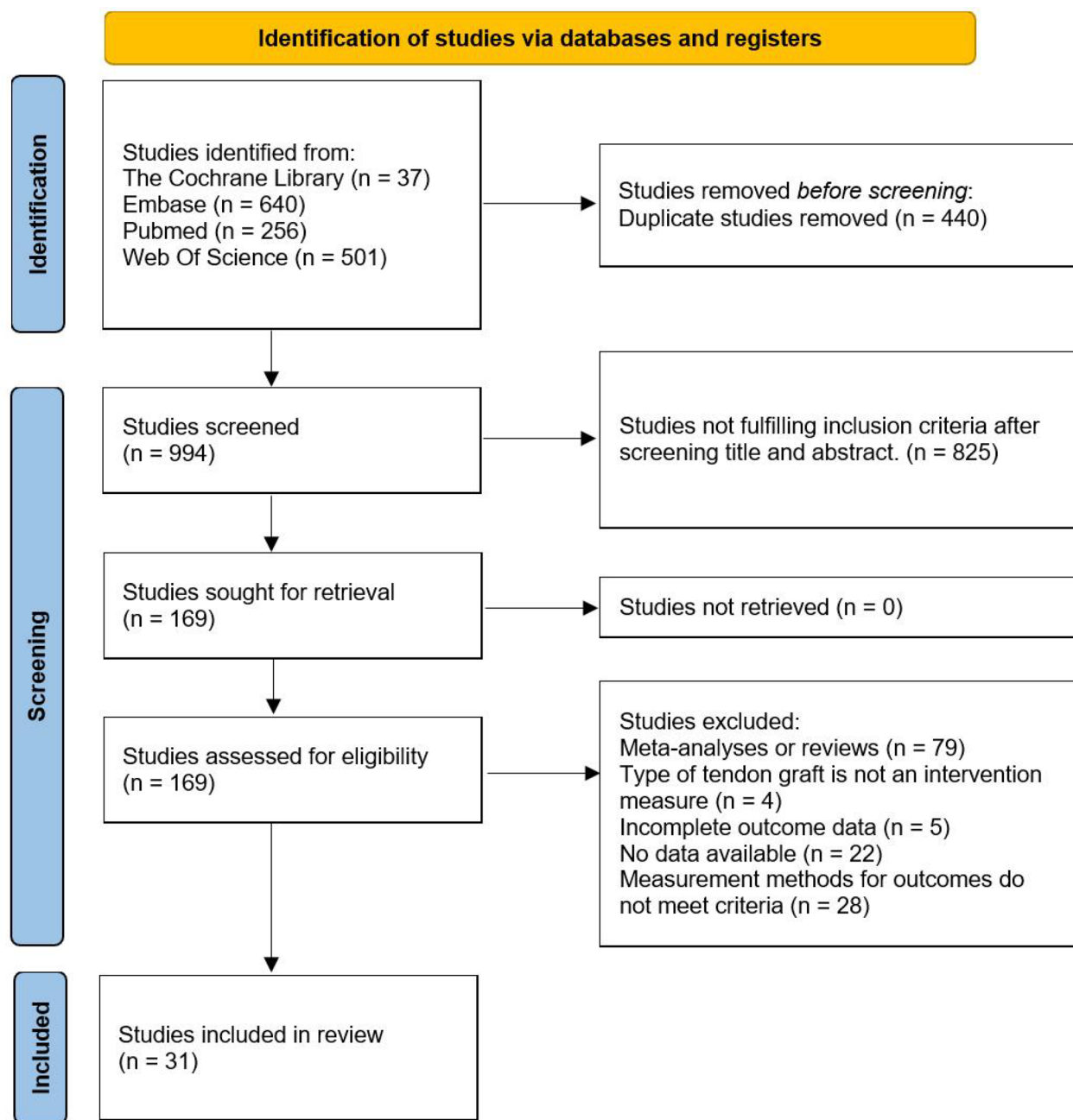


Fig. 1 PRISMA flow diagram for comparison of muscle strength recovery after ACLR using different autografts

studies [54–73], with a total of 2384 patients from all over the world. Interventions are referred to the different autograft types being compared. The distribution of autograft types was as follows: bone-patellar tendon-bone autograft (BPTB autograft, $n = 546$), patellar tendon autograft without a bone block (PT autograft, $n = 51$), four-strand semitendinosus tendon autograft (4S-ST autograft, $n = 299$), five-strand semitendinosus tendon autograft (5S-ST autograft, $n = 18$),

four-strand hamstring tendon autograft using semitendinosus and gracilis (4S-HT autograft, $n = 1120$) and quadriceps tendon autograft (QT autograft, $n = 350$). Male patients comprised 64.2% of the total (excluding 4 studies with missing sex data), and the mean age was 27.0 years (excluding 3 studies with missing age data). The mean follow-up duration was 26.1 months (median, 12.0 months; range, 6–128.4 months). Ten studies reported loss to follow-up, with a total of 292

patients affected. Data extraction results are presented in Supplement Table S2.

Risk of Bias Assessment

The risk of bias (RoB) of the included RCTs was evaluated using the Cochrane RoB tool (Supplementary Figure S1). The independent reviewers' judgments concerning each bias risk item are presented as percentages across all studies, as illustrated in Supplementary Figure S1a. Supplementary Figure S1b presents an overview of the assessments conducted by independent reviewers for each risk of bias item across all studies included in the analysis. Most studies (91%) reported random participant selection, though specific randomization methods were often unclear. Blinding of participants, personnel and outcome assessors was generally not feasible due to differences in surgical procedures, resulting in high risk of performance and detection bias. While 64% of studies adopted adequate allocation concealment, 45% reported incomplete outcome data, and selective reporting risk was unclear. For observational studies, risk of bias was evaluated using RoBANS (Supplementary Table S5). Most studies were retrospective, with high risk of selection bias (85%) and high risk of bias in outcome assessment due to lack of blinding. Confounding variables were adequately controlled in 60% of studies, and standardized

isokinetic measurements minimized exposure bias. Loss to follow-up affected 35% of studies, and selective reporting risk was unclear. Overall, the main biases were related to selection and outcome assessment, whereas measurement reliability and confounding control were generally satisfactory.

Knee Extensor LSI at 60°/s

All Included Studies

Network meta-analysis for knee extensor LSI at 60°/s included 14 studies with 1608 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 48, 52, 53, 55, 58, 61, 63–65, 67, 68, 70, 71] (Fig. 2a). The largest group was 4S-HT autograft ($n=813$). Forest plot analysis presented MDs with 95% CIs among the above-mentioned autograft types, and all detailed comparison results were displayed in Fig. 2b; among these comparisons, except for the one between 4S-HT and 4S-ST, all others showed significant differences: BPTB versus 4S-ST (MD = -6.76, 95% CI -12.10 to -1.42; certainty of evidence, very low in both observational studies and RCTs), QT versus 4S-ST (MD = -10.14, 95% CI -15.44 to -4.85; certainty, very low in both observational studies and RCTs), BPTB versus 4S-HT (MD = -5.32, 95% CI -7.55 to -3.08; certainty, very low in observational studies and low in RCTs), QT versus 4S-HT (MD = -8.70, 95% CI -10.82 to -6.59; certainty, very low in observational studies and low in RCTs), QT versus BPTB (MD = -3.39, 95% CI -6.41 to -0.37)

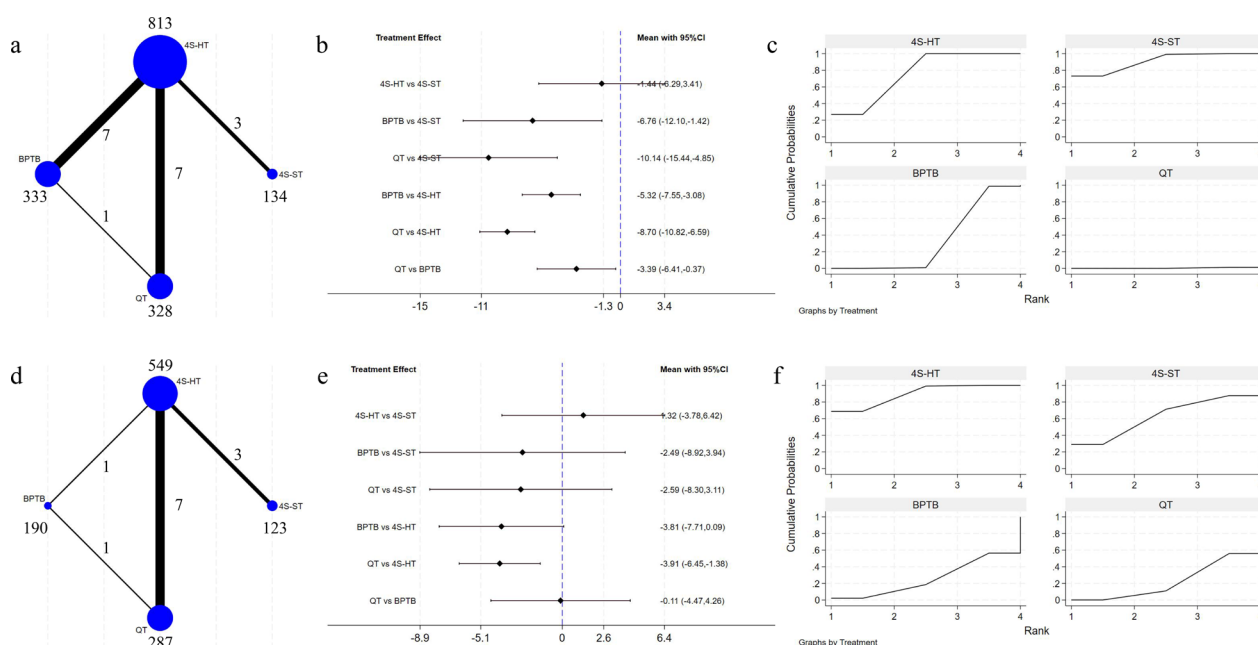


Fig. 2 Network analysis of knee extensor LSI. **a** Network map of knee extensor LSI at 60°/s. Blue dots represent sample size, line thickness indicates the number of comparisons, and numbers indicate the exact sample. **b** Forest plot (mean difference) of knee extensor LSI at 60°/s. **c** Probability rankings of knee extensor LSI at 60°/s. **d** Network map of knee extensor LSI at 180°/s. **e** Forest plot (mean difference) of knee extensor LSI at 180°/s. **f** Probability rankings of knee extensor LSI at 180°/s. Abbreviations: 4S-HT autograft, four-strand hamstring tendon autograft; 4S-ST autograft, four-strand semitendinosus tendon autograft; BPTB autograft, bone–patella tendon–bone autograft; QT autograft, quadriceps tendon autograft

to -6.59 ; certainty, very low in observational studies and low in RCTs), and QT versus BPTB ($MD = -3.39$, 95% CI -6.41 to -0.37 ; certainty, very low in both observational studies and RCTs). SUCRA rankings (Fig. 2c) were 4S-ST 90.8%, 4S-HT 75.6%, BPTB 33.2%, and QT 0.4%. The network showed no significant inconsistency ($P = 0.185$). The results of the GRADE evidence grading were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 6 studies involving 521 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [58, 61, 64, 67, 68, 71] (Supplementary Figure S2a). The largest group was the 4S-HT autograft ($n = 307$). Forest plot analysis (Supplementary Figure S2b) showed that significant differences were observed between BPTB and 4S-HT ($MD = -7.12$, 95% CI $[-14.21, -0.02]$), between QT and 4S-HT ($MD = -10.56$, 95% CI $[-13.50, -7.61]$), and between QT and 4S-ST ($MD = -13.52$, 95% CI $[-23.59, -3.45]$). The SUCRA rankings (Supplementary Figure S2c) were 4S-ST 89.0%, 4S-HT 74.9%, BPTB 30.0%, and QT 6.1%. Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 7 studies with a total of 532 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 48, 52, 53, 58, 67, 68] (Supplementary Figure S2d). The largest group was the 4S-HT autograft ($n = 291$). Forest plot analysis (Supplementary Figure S2e) demonstrated significant differences between 4S-HT and both BPTB ($MD = 6.00$, 95% CI $[1.49, 10.50]$) and QT ($MD = 7.96$, 95% CI $[4.57, 11.35]$). The SUCRA rankings (Supplementary Figure S2f) were 4S-HT 92.3%, 4S-ST 58.4%, BPTB 35.4%, and QT 13.9%. Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with a total of 555 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [48, 55, 63, 65, 70] (Supplementary Figure S2g). The largest group was the 4S-HT autograft ($n = 215$). Forest plot analysis (Supplementary Figure S2h) showed that significant differences existed between BPTB and both 4S-HT ($MD = -4.35$, 95% CI $[-7.14, -1.57]$) and 4S-ST ($MD = -8.03$, 95% CI $[-15.74, -0.32]$). The SUCRA rankings (Supplementary Figure S2i) were 4S-ST 88.5%, 4S-HT 60.0%, QT 44.0%, and BPTB 7.4%. The inconsistency test indicated no significant inconsistency ($P = 0.8840$), confirming a reliable network.

Knee Extensor LSI at $180^\circ/s$

All Included Studies

The network meta-analysis of knee extensor LSI at $180^\circ/s$ (Fig. 2d) included 10 studies with 1149 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 47, 55, 59, 61, 63, 65, 67, 70, 73]. The 4S-HT autograft had the largest sample ($n = 549$). Forest plot analysis (Fig. 2e) showed that a significant difference was observed between QT and 4S-HT ($MD = -3.91$, 95% CI $[-6.45, -1.38]$). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 2f) were 4S-HT 89.3%, 4S-ST 62.6%, BPTB 25.7%, and QT 22.3%. Inconsistency and heterogeneity tests confirmed a reliable network ($P = 0.3015$). The results of the GRADE evidence grading were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 4 studies with 323 patients and 3 autograft types: 4S-ST, 4S-HT and QT [47, 61, 67, 73] (Supplementary Figure S3a), with 4S-HT autograft having the largest sample ($n = 181$). Forest plot analysis (Supplementary Figure S3b) showed a significant difference between QT and 4S-HT ($MD = -8.01$, 95% CI $[-13.00, -3.03]$). SUCRA rankings were 4S-HT 84.0%, 4S-ST 62.5%, and QT 3.5% (Supplementary Figure S3c). No inconsistency assessment was performed due to limited direct comparisons.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 3 studies with 279 patients and 2 autograft types: 4S-HT and QT [44, 67, 70] (Supplementary Figure S3d), with 4S-HT having the largest sample ($n = 155$). Forest plot analysis (Supplementary Figure S3e) indicated that a significant difference existed between QT and 4S-HT ($MD = -5.12$, 95% CI $[-8.38, -1.86]$), and SUCRA rankings were 4S-HT 100% versus QT 0% (Supplementary Figure S3f). No inconsistency assessment was conducted.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with 547 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [55, 59, 63, 65, 70] (Supplementary Figure S3g), with 4S-HT autograft again having the largest sample ($n = 213$). Forest plot analysis (Supplementary Figure S3h) demonstrated the presence of a significant difference between BPTB and 4S-HT ($MD = -3.33$, 95% CI $[-5.74, -0.92]$). SUCRA rankings were 4S-HT 73.9%, QT 63.8%, 4S-ST 50.9%, and BPTB 11.4%

(Supplementary Figure S3i). Inconsistency testing confirmed a reliable network ($P=0.5928$).

Knee Flexor LSI at 60°/s

All Included Studies

The network analysis comparing knee flexor LSI at 60°/s was presented in Fig. 3a. 13 studies with 1477 patients and 4 autograft types (BPTB, 4S-ST, 4S-HT and QT) [44, 48, 52–54, 58, 61, 63–65, 67, 68, 70] were included, with 4S-HT autograft having the largest sample size ($n=749$). Forest plot analysis (Fig. 3b) showed a significant difference between QT and 4S-HT (MD=9.20, 95% CI [2.80, 15.59]). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 3c) were QT 84.9%, 4S-ST 65.2%, BPTB 44.8%, and 4S-HT 5.1%. No inconsistency assessment was performed, and the GRADE evidence grading results were presented in Supplementary Table S6.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 5 studies with 410 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [58, 61, 64, 67, 68] (Supplementary Figure S4a), with 4S-HT as the largest group ($n=235$). No statistically significant difference was observed (Supplementary Figure S4b), and SUCRA rankings were 4S-ST 67.2%, QT 57.0%, BPTB

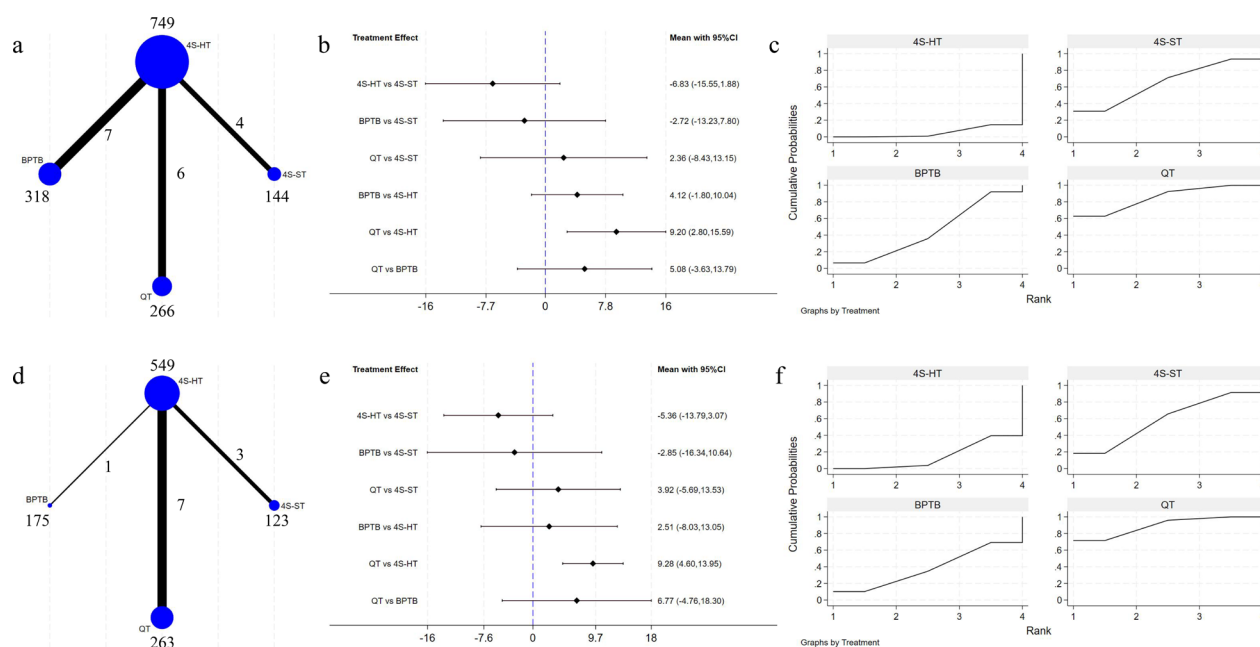
51.3%, and 4S-HT 24.6% (Supplementary Figure S4c). No inconsistency assessment was conducted.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 7 studies with 532 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [44, 52, 53, 58, 67, 68, 70] (Supplementary Figure S4d), with 4S-HT as the largest group ($n=291$). Forest plot analysis (Supplementary Figure S4e) showed significant differences between 4S-HT and both BPTB (MD=−6.20, 95% CI [−11.10, −1.30]) and QT (MD=−10.84, 95% CI [−15.68, −6.01]). SUCRA rankings were QT 93.5%, BPTB 55.7%, 4S-ST 48.4%, and 4S-HT 2.3% (Supplementary Figure S4f). Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 5 studies with 535 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [48, 54, 63, 65, 70] (Supplementary Figure S4g), with 4S-HT as the largest group ($n=225$). No statistically significant difference was observed (Supplementary Figure S4h), and SUCRA rankings were QT 76.8%, 4S-ST 58.4%, BPTB 35.7%, and 4S-HT 29.1% (Supplementary Figure S4i). No inconsistency assessment was performed due to limited direct comparisons.



Knee Flexor LSI at 180°/s

All Included Studies

The network meta-analysis of knee flexor LSI at 180°/s is summarized as follows. 9 studies with 1109 patients and 4 autograft types (BPTB, 4S-ST, 4S-HT and QT) [44, 47, 59, 61, 63, 65, 67, 70, 73] (Fig. 3d) were included, with 4S-HT autograft having the largest sample size ($n=549$). Forest plot analysis (Fig. 3e) showed a significant difference between QT and 4S-HT (MD=9.28, 95% CI [4.60, 13.95]). The certainty of evidence for this comparison was very low in observational studies and low in RCTs. SUCRA rankings (Fig. 3f) were QT 89.2%, 4S-ST 58.4%, BPTB 38.0%, and 4S-HT 14.4%. No inconsistency assessment was performed, and Supplementary Table S6 contains the results of the GRADE evidence grading.

Subgroup 1: Follow-Up ≤ 6 months

Subgroup 1 included 4 studies with 323 patients and 3 autograft types: 4S-ST, 4S-HT and QT [47, 61, 67, 70] (Supplementary Figure S5a), with 4S-HT as the largest group ($n=181$). Forest plot analysis (Supplementary Figure S5b) showed that no statistically significant difference was observed. SUCRA rankings were 4S-ST 74.4%, QT 59.4%, and 4S-HT 16.2% (Supplementary Figure S5c). No inconsistency assessment was conducted.

Subgroup 2: Follow-Up 6–12 months

Subgroup 2 included 3 studies with 279 patients and 2 autograft types: 4S-HT and QT [44, 67, 70] (Supplementary Figure S5d), with 4S-HT as the largest group ($n=155$). Forest plot analysis (Supplementary Figure S5e) showed that a significant difference existed between QT and 4S-HT (MD=13.93, 95% CI [9.79, 18.06]). SUCRA rankings were QT 100% and 4S-HT 0% (Supplementary Figure S5f). Inconsistency assessment was not performed due to the lack of direct comparisons.

Subgroup 3: Follow-Up ≥ 18 months

Subgroup 3 included 4 studies with 507 patients and 4 autograft types: BPTB, 4S-ST, 4S-HT and QT [59, 63, 65, 70] (Supplementary Figure S5g), with 4S-HT as the largest group ($n=213$). No statistically significant difference was observed (Supplementary Figure S5h). SUCRA rankings were QT 75.5%, 4S-ST 51.0%, BPTB 47.6%, and 4S-HT 25.8% (Supplementary Figure S5i). No inconsistency assessment was performed due to limited direct comparisons.

Comparisons of Secondary Outcomes

The comparison of peak torque across different angular velocities involved 6 autograft types (BPTB, PT, 4S-ST, 5S-ST, 4S-HT and QT). Forest plots (Supplementary Figures S6–S9) showed no statistically significant

differences in knee extensor or flexor peak torque among the autografts at 60°/s or 180°/s. Notably, some autograft types had small sample sizes (5S-ST, 18 patients; PT, 30 patients).

Postoperative muscle strength recovery assessed by H/Q ratio was analyzed for 4 autograft types at 60°/s and 180°/s. Forest plot analysis (Supplementary Figures S10–S11) indicated that significant differences in H/Q ratio were observed between QT and 4S-HT at both 60°/s (MD=12.79, 95% CI [3.29, 22.29]) and 180°/s (MD=12.99, 95% CI [0.26, 25.73]), and between 4S-ST and 4S-HT at 180°/s (MD=7.99, 95% CI [2.58, 13.41]). Good network consistency was observed at 60°/s ($P=0.2064$).

Side-to-side deficits in extensor and flexor peak torque were compared across 5 autograft types (BPTB, PT, 4S-ST, 4S-HT and QT). Among the findings of this comparison, a significant difference in extensor peak torque side-to-side deficits was observed between BPTB and 4S-HT at 180°/s (MD=3.09, 95% CI [0.71, 5.47]). For flexor peak torque side-to-side deficits at 180°/s, significant differences were noted between QT and each of 4S-HT (MD=−11.90, 95% CI [−20.90, −2.89]), 4S-ST (MD=−9.00, 95% CI [−14.34, −3.66]), and BPTB (MD=−12.90, 95% CI [−24.27, −1.52]). No significant differences were observed at 60°/s. Additionally, the small sample of PT autograft ($n=21$) should be noted (Supplementary Figures S12–S15).

Assessment of Publication Bias

Funnel plots were generated in Stata to assess potential publication bias (Supplementary Figure S16). Plots for knee flexion and extension at different angular velocities showed no significant asymmetry, indicating minimal publication bias in this study.

Discussion

This study provides a comprehensive evaluation of comparisons in knee extensor and flexor muscle strength after ACLR using different autografts. Compared with previous meta-analyses [7, 8], the present meta-analysis included a larger number of studies and autografts with indirect comparisons. By synthesizing data from 6 autograft types, 31 studies and 2384 patients, we assessed postoperative isokinetic muscle strength at 60°/s and 180°/s, thereby offering a holistic perspective on graft-specific recovery of muscle strength. Subgroup analyses by follow-up duration (short-term, ≤ 6 months; mid-term, 6–12 months; long-term, ≥ 18 months) were conducted to explore temporal differences in recovery. In summary, across both angular velocities (60°/s and 180°/s), the 4S-HT autograft showed superior knee extensor strength recovery, whereas the QT autograft

consistently showed the greatest potential in flexor strength recovery. The 4S-ST autograft occupied an intermediate position, exhibiting advantages in certain long-term follow-ups and showing greater knee flexor strength LSI at 60°/s compared with the 4S-HT autograft. Additionally, the BPTB autograft primarily demonstrated greater knee extensor strength recovery than the QT autograft at 60°/s. However, given the very low to low certainty of evidence, these findings should be interpreted with caution.

Knee Extensor LSI at 60°/s and 180°/s

For knee extensor LSI at 60°/s, forest plots and SUCRA rankings suggested that 4S-ST and 4S-HT autografts may confer relative advantages over BPTB and QT autografts across the overall follow-up period. In contrast, Johnston et al. [7] and Herbawi et al. [8] reported that the 4S-HT autograft primarily demonstrated superiority over BPTB and QT autografts in knee extensor strength recovery when follow-up was ≤ 6 months, but no difference was observed between the 4S-HT autograft and BPTB and QT autografts when follow-up was ≥ 12 months [8]. This discrepancy may reflect the combined direct and indirect evidence used in the present network meta-analysis, whereas previous studies relied mainly on direct pairwise comparisons. Specifically, our analysis included both direct 4S-HT versus QT evidence and indirect evidence through BPTB, broadening the evidence base for the 4S-HT versus QT estimate. The discrepancy may also be explained by smaller sample sizes in previous reviews, which included only two studies for each relevant subgroup (170 and 142 patients), limiting statistical power. Although no significant difference was observed between 4S-ST and 4S-HT autografts, the certainty of evidence for this comparison was very low, reducing confidence in the finding. In the overall comparison, the BPTB autograft demonstrated greater knee extensor strength recovery than the QT autograft, while in subgroup analyses, smaller sample sizes may have widened confidence intervals and reduced statistical significance.

In the overall comparison of knee extensor LSI at 180°/s, forest plots and SUCRA rankings indicated that the 4S-HT autograft consistently outperformed other grafts across all follow-up periods. It ranked first in the overall, short-term, mid-term, and long-term analyses, showing stable recovery of knee extensor strength and demonstrating significant superiority over the QT autograft in the overall, short-term and mid-term follow-ups, as well as over the BPTB autograft in the long-term follow-up. These findings are consistent with previous meta-analysis [9].

There were differences in knee extensor strength recovery following QT autograft among different follow-up

subgroups. Although QT autografts performed less favorably in short- and mid-term analyses, their ranking improved substantially at long-term follow-up, rising to second at 180°/s and third at 60°/s and surpassing BPTB and 4S-ST autografts in this subgroup. This pattern is consistent with studies showing greater knee extensor-strength recovery with HT than QT autografts at 6 months but no significant difference among QT, BPTB and HT autografts at 24 months [8]. These patterns may reflect time-dependent recovery, but they may also be affected by the small number of long-term studies, limited sample sizes, non-standardized rehabilitation protocols and incomplete intervention networks (the short-term phase lacked BPTB autografts, and the mid-term phase lacked both 4S-ST and BPTB autografts). Moreover, the certainty of evidence for QT versus BPTB at 60°/s and QT versus 4S-ST at 180°/s was very low, limiting confidence in these subgroup findings. Future studies should follow a cohort prospectively to analyze the temporal trajectory of knee extensor strength recovery in patients with QT autograft for ACLR.

Knee Flexor LSI at 60°/s and 180°/s

In the overall comparison of the LSI for knee flexor strength at 60°/s after ACLR, the QT autograft had the highest SUCRA values in the overall analysis and most subgroups, consistent with previous research [7, 8]. However, no statistically significant intergroup differences were observed in the short-term (follow-up ≤ 6 months) and long-term (follow-up ≥ 18 months) follow-ups; only in the mid-term (follow-up 6–12 months) did QT and BPTB autografts show a trend of superiority over the 4S-HT autograft. Herbawi et al. [8] reported significantly better short-term outcomes with the QT autograft compared with 4S-HT, whereas our results did not confirm this. This discrepancy may be partly explained by the small sample size of the QT group (approximately half the size of the HT group) in Johnston et al. [67], thereby introducing a bias toward the larger comparator group. At long-term follow-up, knee flexor strength recovery with QT was comparable to that with 4S-HT and BPTB, which aligns with Herbawi et al. [8]. The 4S-HT autograft consistently demonstrated relative disadvantages in knee flexor strength recovery, likely due to the reduced hamstring cross-sectional area following tendon harvest [74]. Supporting this, Girdwood et al. [6] reported a 10–15% greater knee flexor strength deficit in hamstring tendon graft recipients compared with other graft types. In the present study, no significant difference was observed between the 4S-ST and 4S-HT autograft in knee flexor strength recovery across the overall sample and all subgroups, while the SUCRA plot indicated a high probability that the 4S-ST autograft was superior to the 4S-HT

autograft in this regard, consistent with prior research that has indicated superior knee flexor strength recovery with 4S-ST over 4S-HT [10, 75, 76]. The absence of statistical significance may reflect methodological differences, as previous studies used conventional pairwise meta-analysis whereas the present study incorporated evidence from multiple comparative pairs. Similarly, no significant difference was observed between BPTB and QT autografts, although SUCRA rankings suggested a high probability that QT was superior for knee flexor-strength recovery.

In the overall comparison of LSI for knee flexor strength at 180°/s after ACLR, results were largely consistent with those observed at 60°/s, except for the absence of data on certain interventions—specifically, the comparison in short-term (Follow-up ≤ 6 Months) lacked data on BPTB autograft, while the comparison in mid-term (Follow-up 6–12 Months) lacked data on both BPTB and 4S-ST autograft.

Comparisons of Secondary Outcomes

In the analysis of H/Q ratio, a higher H/Q ratio for QT autografts indicated better knee flexor-strength recovery than 4S-HT autografts at both 60°/s and 180°/s, consistent with the findings of Salas et al. [77]. Additionally, side-to-side deficit analyses at 180°/s suggested that the QT autograft may outperform the 4S-HT, 4S-ST, and BPTB autograft in knee flexor strength recovery. These results align with the primary outcome measure (LSI). By contrast, peak torque comparisons revealed no significant differences among autografts, likely due to the lack of adjustment for preoperative torque differences, which were not reported in the included studies. Additionally, residual confounding may also arise from unadjusted variables including activity level, body mass index, and sex. Moreover, the extremely small sample sizes for the PT and 5S-ST autograft rendered their comparative outcomes highly unreliable.

Clinical implications

The present findings may provide several cautious clinical implications for graft selection and rehabilitation planning after ACLR. Although graft choice should primarily be individualized according to patient age, activity level, sport-specific demands, anatomical considerations, surgeon experience, and patient preference [78–80], the observed graft-specific patterns of muscle strength recovery may serve as supplementary information during shared decision-making. The relatively favorable knee extensor strength recovery observed with the 4S-HT and 4S-ST autografts suggests that

these grafts may be considered when early or sustained quadriceps strength recovery is a particular clinical priority. Conversely, the QT autograft showed a tendency toward better knee flexor strength recovery, which may be relevant for patients in whom preservation of hamstring function is important, such as those participating in sports requiring repeated sprinting, cutting, or dynamic knee stabilization. However, these findings should not be interpreted as evidence that one graft is universally superior to another. From a rehabilitation perspective, the results highlight the need for graft-specific monitoring and targeted strength training after ACLR. Patients receiving 4S-HT or 4S-ST autografts may require particular attention to knee flexor strength restoration. In contrast, patients receiving QT or BPTB autografts may require careful monitoring of quadriceps strength recovery, especially during the early and mid-term postoperative periods.

Limitations

Several limitations should be acknowledged. First, variability in rehabilitation protocols may have contributed to differences among follow-up subgroups. Rehabilitation is a critical determinant of muscle strength recovery; however, insufficient reporting prevented assessment of rehabilitation-related factors. Previous research indicates that approximately 40% of patients receive supervised rehabilitation for less than 3 months after ACLR [81], and only 40% continue rehabilitation beyond 9 months [82], suggesting that many patients may receive inadequate rehabilitation. Additional RCTs with standardized rehabilitation protocols are therefore needed to strengthen the credibility of future evidence. At present, however, no consensus exists regarding the optimal dose, duration or components of post-ACLR rehabilitation [83]. Second, because few RCTs assessed postoperative knee muscle strength, observational studies were included to support the network meta-analysis. Although methods such as sample matching were used to reduce confounding, the GRADE certainty of evidence remained very low, highlighting the need for more high-quality RCTs. Third, most trials compared BPTB and 4S-HT autografts, leaving other intervention groups with small samples and many absent direct comparisons. Consequently, closed-loop networks were incomplete, limiting statistical assessment of consistency among indirect comparisons [84]. Several effect estimates also had wide confidence intervals, indicating low precision and possible variability in true effects. These limitations mean that the results should be interpreted cautiously, and further well-designed

head-to-head RCTs with standardized rehabilitation protocols are warranted.

Conclusions

Overall, integrating findings from the pooled and subgroup analyses, this network meta-analysis suggests that among commonly used autografts for ACLR, the 4S-HT autograft was associated with greater knee extensor strength LSI, whereas the QT autograft had the most consistent advantage in knee flexor strength LSI. The 4S-ST autograft ranked between these grafts, but may provide greater knee flexor strength LSI than the 4S-HT autograft. Additionally, the BPTB autograft was associated with greater knee extensor strength LSI than the QT autograft. However, given the very low to low certainty of the available evidence, these findings should be interpreted with caution, and further high-quality, large-scale randomized controlled trials are therefore warranted to validate these results and guide optimal graft selection in clinical practice.

Abbreviations

ACLR	Anterior cruciate ligament reconstruction
LSI	Limb symmetry index
RTS	Return to sport
RCT	Randomized controlled trial
BPTB	Bone-patella tendon-bone
PT	Patella tendon
4S-ST	Four-strand semitendinosus tendon
5S-ST	Five-strand semitendinosus tendon
4S-HT	Four-strand hamstring tendon
QT	Quadriceps tendon
H/Q	Hamstring-to-quadriceps
RoB	Risk of bias
GRADE	Grading of Recommendations Assessment, Development and Evaluation
SUCRA	Surface under the cumulative ranking curve
MD	Mean difference
CI	Confidence interval
PRISMA-NMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses-Network Meta-Analyses
AMSTAR	Assessing the Methodological Quality of Systematic Reviews
RoBANS	Risk of Bias Assessment Tool for Non-randomized Studies
IKDC	International Knee Documentation Committee
PRISM	Pediatric Research in Sports Medicine

Supplementary Information

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Additional file 1.

Additional file 2.

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Author Contributions

C.L. and W.L. Wrote original draft. W.X. and Y.L. conceptualized this article and revised the draft. Z.H. and Q.B. suggested revisions. C.L., W.L. and X.L. undertook the literature review and extracted the data. C.L., W.L. and D.Y. coded the statistical analysis, figures, and appendix. Quality assessment of included studies by

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Data Availability

All data relevant to the study are included in the article or uploaded as supplementary information.

Declarations

Ethics Approval and Consent to Participate

Not applicable.

Informed Consent

Not applicable.

Equity, Diversity and Inclusion Statement

Both randomized controlled trials (RCTs) and observational studies were eligible, without restriction by country or region, thereby encompassing diverse racial and ethnic groups. Our author team includes senior and junior scholars from Asia. Currently, our team is all male, which reflects the existing composition rather than deliberate exclusion. The study did not examine the impact of race, ethnicity, or socioeconomic status on outcomes.

Provenance and Peer Review

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Competing Interests

The authors declare that they have no competing interests.

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